

International Mathematical Olympiad 2024

Hong Kong Team Selection Test 1

22 Aug 2023

Problem 1. Let n be a positive integer larger than 1, and let $a_0, a_1, \dots, a_{n-1}$ be integers. It is known that the equation

$$x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x + a_0 = 0$$

has n pairwise relatively prime integer roots. Prove that a_0 and a_1 are relatively prime.

Problem 2. We say that a real number is ‘almost an integer’ if it differs from an integer by at most 0.1. For example, 2023, -2023.9 and 2023.0822 are almost integers. Show that among any 10 real numbers, there exist two different real numbers whose difference is almost an integer.

Problem 3. Given $\triangle ABC$ with $AB < AC$, let AD be the bisector of $\angle BAC$ with D on the side BC . Let Γ be a circle passing through A and D which is tangent to BC at D . Suppose Γ cuts the side AB again at $E \neq A$. The tangent to the circumcircle of $\triangle BDE$ at D intersects Γ again at $F \neq D$. Let P be the intersection point of the segments EF and AC . Prove that PD is perpendicular to BC .

Problem 4. After expanding the polynomial $(1 + x + y)^{2023}$ and collecting like terms, we obtain the expression

$$a_0 + a_1x + a_2y + a_3x^2 + a_4xy + a_5y^2 + \dots + a_ky^{2023}.$$

Find the total number of a_i 's which are divisible by 5.

Problem 5. Find all nonnegative real numbers a, b, c such that

$$\frac{4a + 9b + 25c}{2a + 3b + 5c} + \frac{4b + 9c + 25a}{2b + 3c + 5a} + \frac{4c + 9a + 25b}{2c + 3a + 5b} = 10.$$

Problem 6. Let Ω be the incircle of $\triangle ABC$. There are two smaller circles ω_1 and ω_2 inside $\triangle ABC$. The circle ω_1 is tangent to Ω at P , tangent to BC at D , and also tangent to AB . The circle ω_2 is tangent to Ω at Q , tangent to BC at E , and also tangent to AC . Prove that D, E, Q, P are concyclic.

Time: 3 hours

Each problem is worth 7 points

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問題 1. 設 n 是一個大於 1 的正整數，並設 $a_0, a_1, \dots, a_{n-1}$ 為整數。已知方程

$$x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x + a_0 = 0$$

有 n 個兩兩互質的整數根。求證： a_0 和 a_1 互質。

問題 2. 對於一個實數，若存在一個整數與它的差不超過 0.1，則我們稱它「幾乎是整數」，例如 2023、 -2023.9 及 2023.0822 都幾乎是整數。求證：在任何 10 個實數當中，必定存在兩個不同的實數，它們之差幾乎是整數。

問題 3. 在 $\triangle ABC$ 中， $AB < AC$ ，設 AD 是 $\angle BAC$ 的平分線，其中 D 在邊 BC 上。設 Γ 是穿過 A 和 D 的圓形，並與 BC 相切於 D 。假設 Γ 與邊 AB 再相交於 $E \neq A$ 。在點 D 對於 $\triangle BDE$ 的外接圓的切線與 Γ 再相交於 $F \neq D$ 。設 P 是線段 EF 和 AC 的交點。求證： PD 與 BC 垂直。

問題 4. 將多項式 $(1 + x + y)^{2023}$ 展開並將同類項合併後，可得到展開式

$$a_0 + a_1x + a_2y + a_3x^2 + a_4xy + a_5y^2 + \dots + a_ky^{2023}。$$

求能被 5 整除的 a_i 的個數。

問題 5. 求所有非負實數 a, b, c ，使得

$$\frac{4a + 9b + 25c}{2a + 3b + 5c} + \frac{4b + 9c + 25a}{2b + 3c + 5a} + \frac{4c + 9a + 25b}{2c + 3a + 5b} = 10。$$

問題 6. 設 Ω 是 $\triangle ABC$ 的內切圓。 $\triangle ABC$ 內有兩個較小的圓形 ω_1 和 ω_2 。圓 ω_1 與 Ω 相切於 P ，與 BC 相切於 D ，並與 AB 相切；圓 ω_2 與 Ω 相切於 Q ，與 BC 相切於 E ，並與 AC 相切。求證： D, E, Q, P 共圓。

考試時間：3 小時
每題 7 分

Solutions

1. Let $b_1, b_2, \dots, b_n$ be the integer roots. By Vieta's theorem, we have

$$b_1 b_2 \cdots b_n = (-1)^n a_0, \quad (1)$$

$$b_1 b_2 \cdots b_{n-1} + b_1 b_2 \cdots b_{n-2} b_n + \cdots + b_2 b_3 \cdots b_n = (-1)^{n-1} a_1. \quad (2)$$

Suppose on the contrary that a_0 and a_1 share a common prime factor p . By equation (1), we have $p \mid b_1 b_2 \cdots b_n$. WLOG assume $p \mid b_1$. Then in equation (2), p divides each of the first $n-1$ terms on the left and divides the term on the right. Therefore, we must have $p \mid b_2 b_3 \cdots b_n$. This implies one of $b_2, b_3, \dots, b_n$ is divisible by p , contradicting the fact that each of them is relatively prime to b_1 . Therefore, a_0 and a_1 must be relatively prime.

2. Let $x_1, x_2, \dots, x_{10}$ be the real numbers. For each j , we write $x_j = a_j + b_j$ where a_j is the largest integer not exceeding x_j . Note that we must have $0 \leq b_j < 1$. WLOG we may assume $b_1 \leq b_2 \leq \cdots \leq b_{10}$.

- If $b_{j+1} - b_j \leq 0.1$ for some $j = 1, 2, \dots, 9$, then $x_{j+1} - x_j$ is almost an integer since

$$(x_{j+1} - x_j) - (a_{j+1} - a_j) = b_{j+1} - b_j \in [0, 0.1],$$

where $a_{j+1} - a_j$ is an integer.

- If $b_{j+1} - b_j > 0.1$ for all $j = 1, 2, \dots, 9$, then $b_{10} - b_1 > 0.9$, and hence $x_1 - x_{10}$ is almost an integer since

$$(x_1 - x_{10}) - (a_1 - a_{10} - 1) = 1 - (b_{10} - b_1) \in (0, 0.1),$$

where $a_1 - a_{10} - 1$ is an integer.

Therefore, in any case, we can find two different real numbers whose difference is almost an integer.

3. Let $Q \neq A$ be the second intersection point of Γ and AC . As

$$\angle EDB = \angle EAD = \angle DAQ = \angle DEQ,$$

we have $EQ \parallel BC$. Also, we have $AF \parallel BC$ since

$$\angle EAF = 180^\circ - \angle FDE = 180^\circ - \angle DBE.$$

This shows $EQ \parallel AF$, and hence $AEQF$ is an isosceles trapezoid. Thus, the intersection point P of the diagonals lies on the perpendicular bisector of EQ . Note that D also lies on the perpendicular bisector of EQ because AD bisects $\angle EAQ$ and A, E, D, Q are concyclic. Therefore, $PD \perp EQ$, which implies $PD \perp BC$.

5. By the Cauchy-Schwarz inequality, we have

$$(a + b + c)(4a + 9b + 25c) \geq (2a + 3b + 5c)^2.$$

This implies $\frac{4a + 9b + 25c}{2a + 3b + 5c} \geq \frac{2a + 3b + 5c}{a + b + c}$. Adding similar inequalities, we obtain

$$\begin{aligned} & \frac{4a + 9b + 25c}{2a + 3b + 5c} + \frac{4b + 9c + 25a}{2b + 3c + 5a} + \frac{4c + 9a + 25b}{2c + 3a + 5b} \\ & \geq \frac{2a + 3b + 5c}{a + b + c} + \frac{2b + 3c + 5a}{b + c + a} + \frac{2c + 3a + 5b}{c + a + b} \\ & = \frac{10a + 10b + 10c}{a + b + c} \\ & = 10. \end{aligned}$$

By the given condition, the equality should hold. This means we need

$$a : b : c = 4a : 9b : 25c = 25a : 4b : 9c = 9a : 25b : 4c,$$

which means two of a, b, c are zero, and the remaining can be any positive real number.

6. Let S be the point on Ω different from the contact point of Ω and BC such that its tangent to Ω is parallel to BC . Then the tangent at D to ω_1 is parallel to the tangent at S to Ω . By considering the homothety with centre P mapping ω_1 to Ω , we see that D, P, S are collinear. Similarly, E, Q, S are collinear. Let X be a point on the tangent at S to Ω as shown. Then we have

$$\angle SQP = \angle PSX = \angle EDP.$$

This implies D, E, Q, P are concyclic.

